

2 METER 1/2-WAVE J-POLE

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VHF-FM Antenna Made From 450-Ohm LADDER LINE

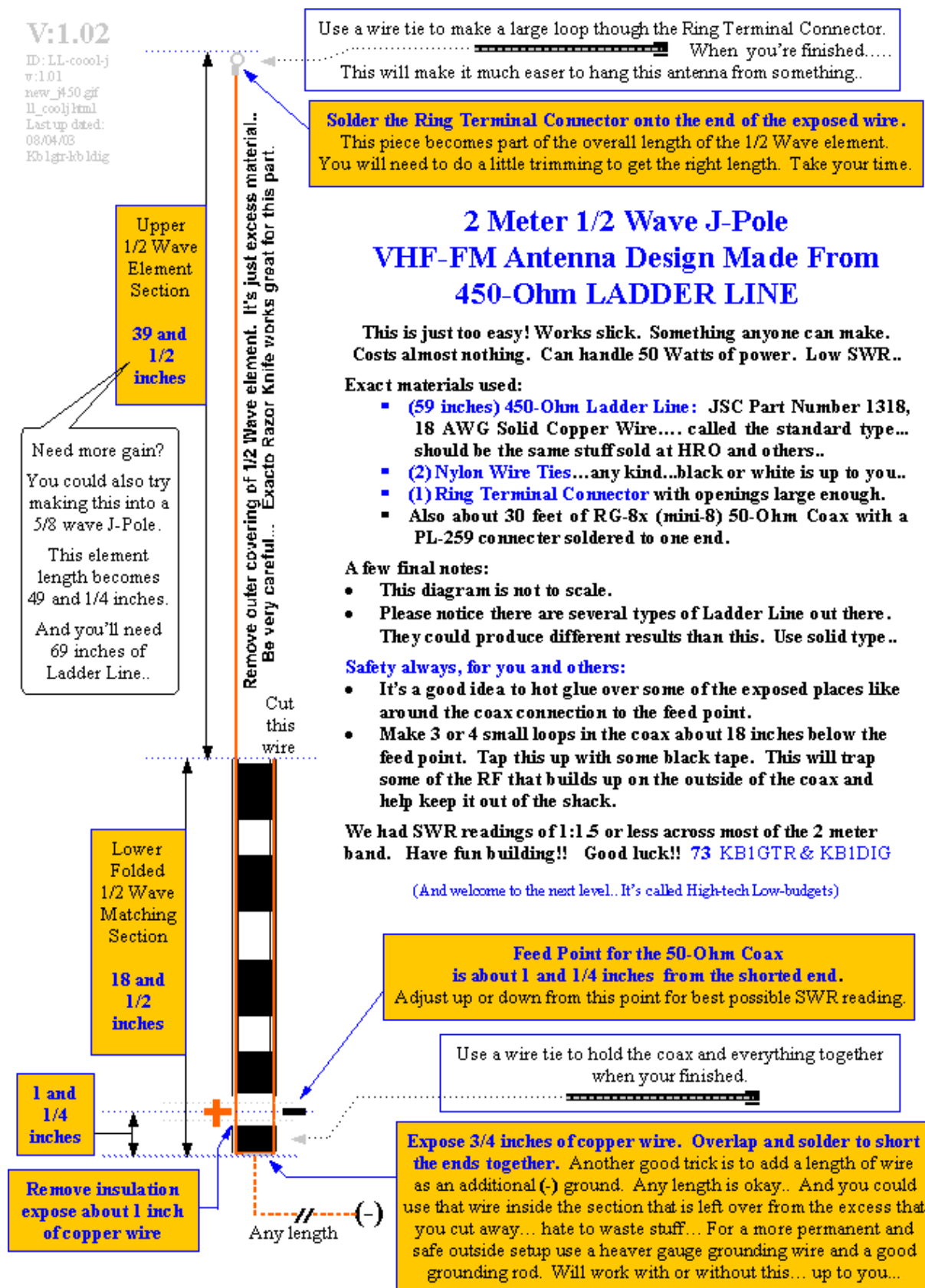
Great first time J-pole project. Covers most of the basics.

This small and thin design also makes it good for several stealth applications.

And.....We've nicknamed this design the "LL-Cool-J"

V:1.02

ID: LL-cool-j
v:1.01
new_j450.gif
ll_coolj.html
Last up dated:
08/04/03
Kb1gr-kb1dig



Performing SWR Testing:

Need a temporary setup for suspending the antenna? Try this proven method.

**Black PVC Electrical Tape
About 3" and Sticky side up**

About half way

Tip of 1/2-wave top element

Place element in center and overlap tape



Punch a hole in the tape to hang antenna



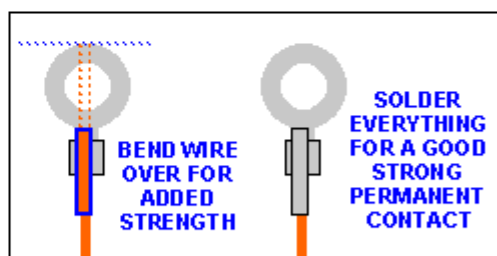
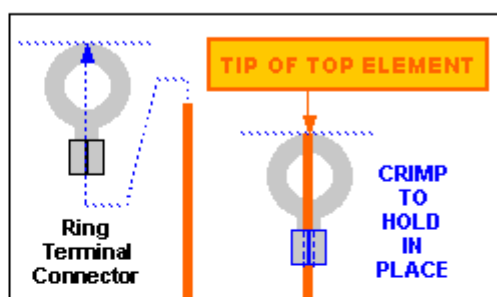
Use something nonconductive like kite string

If you carefully pull the tape apart you may be able to use this assembly over a few times before having to make a new one.

Glue residues are easily removed from the copper element with rubbing alcohol before soldering the Ring Terminal Connector.

Final Assembly Suggestion:

After making all SWR adjustments it's now time to add the Ring Terminal Connector.

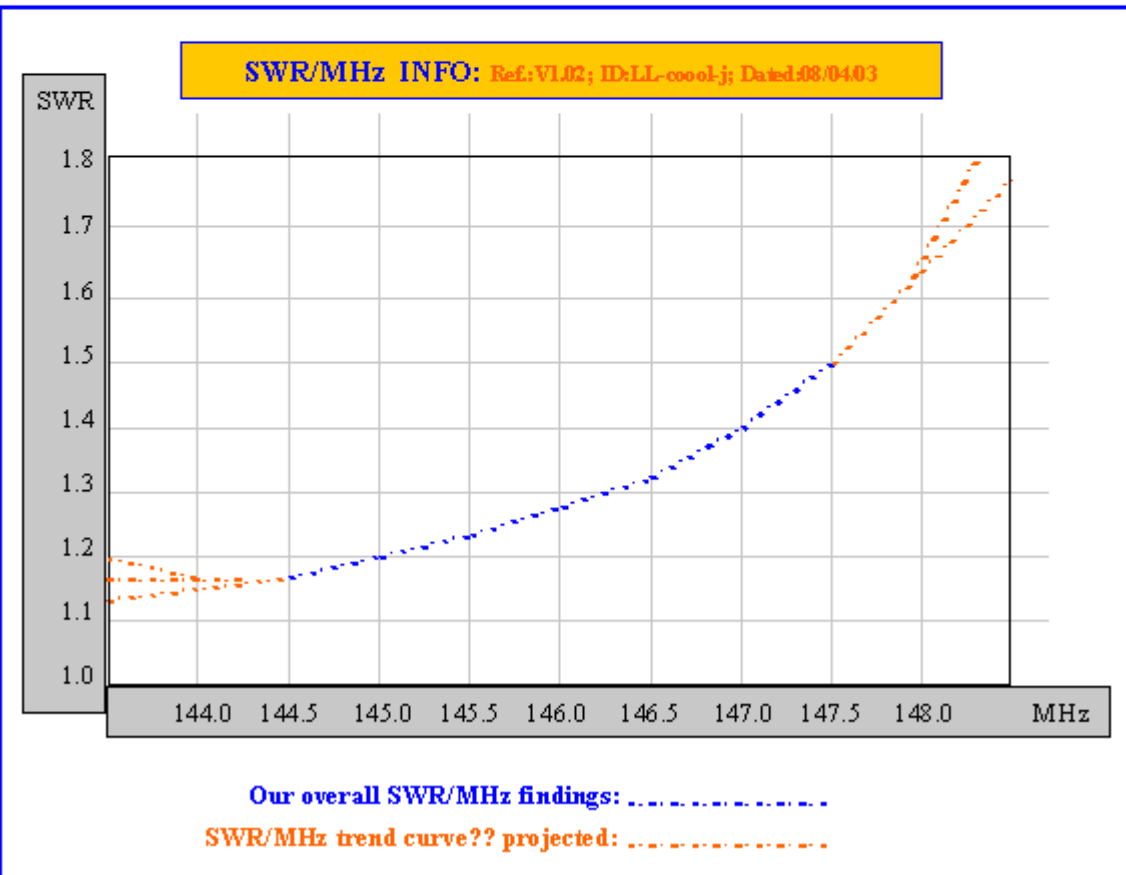


Not much chance of this coming apart under normal circumstances... This antenna is now ready for you to add the wire tie.... Remember to recheck your SWR readings. It should not have changed much at all.

Before you finalize please look closely at your overall SWR/MHz findings..

We were looking for SWR test results of 1.5:1 or less, between 144.500 MHz and 147.500 MHz.

At this point we stopped adjusting. This should meet all of our personal requirements for local 2-meters. Especially good for 2-meter packet. But also great for the type of simplex and repeater work we like to do. The elements ending up a little long is not a bad thing. **Try some extra fine tuning.** As you can see from the **SWR/MHz INFO** below there is still some room for a little tinkering. Something to think about in addition to the other SWR adjustment method at the coax feed point.. You could also trim down the element lengths a little at a time. Do this first before soldering on the Ring Terminal Connector. If you decide to go this route, start by trimming 1/8" off the tip of the "lower matching section element" and then about 3/8" to 1/2" off the "upper 1/2-wave element". This all may need to be done a few times before completion of the assembly... **Remember optimized results will take time.** It all depends on what you're looking for. So, take little steps at a time.. Give it some thought and do the experiments.. Trial and error is how we all learn and is the greatest challenge of extra fine tuning.



Technical Data:

Velocity Factors Estimated: Open Wire@ 98% (insulation removed; as used in upper element section)

Insulated 450-Ohm Ladder Line@ 92% (as used in lower matching section)

Element Length Predictions:

1/2 Wavelength @ 98% = 39.63" +/- 1% (146.000MHz @ 40.44" in free space)

1/4 Wavelength @ 92% = 18.60" +/- 1% (146.000MHz @ 20.22" in free space)

Lower Folded 1/2 Wave Matching Section = 1/4 Wavelength in this calculation

More information to help you build this J-pole antenna for other frequencies:

5/8 Wavelength Formulas:

"615" divided by the frequency in Megahertz (this # is in feet)

(and multiplied by 12 inches) = the free space 5/8 Wavelength in inches

(and then multiplied by the Velocity Factor %) = electrical 5/8 Wavelength in inches

or

$615/\text{MHz} \times 12 \times \text{VF} = \text{electrical } 5/8 \text{ Wavelength in inches}$

1/2 Wavelength Formulas:

"492" divided by the frequency in Megahertz (this # is in feet)

(and multiplied by 12 inches) = the free space 1/2 Wavelength in inches

(and then multiplied by the Velocity Factor %) = electrical 1/2 Wavelength in inches

or

$492/\text{MHz} \times 12 \times \text{VF} = \text{electrical } 1/2 \text{ Wavelength in inches}$

1/4 Wavelength Formulas:

"246" divided by the frequency in Megahertz (this # is in feet)

(and multiplied by 12 inches) = the free space 1/4 Wavelength in inches

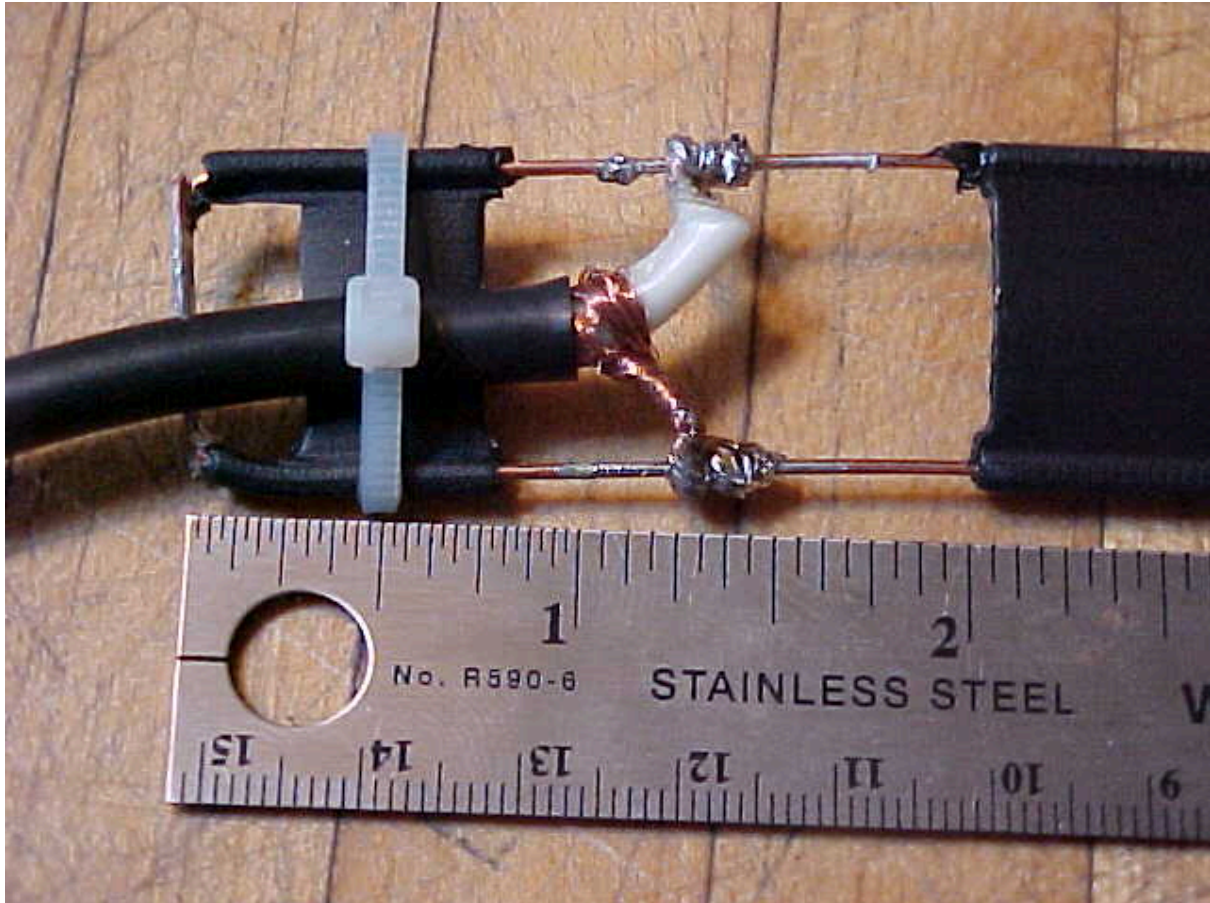
(and then multiplied by the Velocity Factor %) = electrical 1/4 Wavelength in inches

or

$246/\text{MHz} \times 12 \times \text{VF} = \text{electrical } 1/4 \text{ Wavelength in inches}$

As you can see. Had to make a few adjustments.

Re-heating and moving the feed point can be a little difficult. The next step is to cover everything exposed with the hot glue. Also, it's a good idea to get some of that glue in between the wire tie and the coax. This will lock everything into place and help keep the weather out. Fewer problems.. Less headaches later...



This decoupling loop is sometimes mistaken for a matching balun.

We'll just call it a choke and not put everyone to sleep with a long discussion of antenna theory. Just a few points on this topic.. Adding a choke like this is a good habit to get into. Another idea for this is one of those plastic snap-on ferrite cores. They come in different sizes depending on the size coax used. Either way is up to you. This J-pole is a balanced antenna going into an unbalanced feed line and can cause RF to get back into the Rig and into the shack. Save your equipment and try to protect yourself..

This is easy to do.. So, take this extra step.. Please be safe..



A few sample Standard Operating Procedures.

Surprisingly, the 18 gauge wire in this ladder line is very rugged. It's good to keep some open space between the tip of the antenna and whatever you decide to hang it from. Try not to place the antenna close to any metal objects as well. We hang ours from some string thrown over a tree branch. Good to get the antenna at least 10 feet off the ground. And, remember height determines range at 2 meters. The gain for a 1/2 wave J-pole is approximately 2.4 dbi over a 1/4 wave whip. Not bad for only a few bucks.



Construction Improvements & Modification Ideas			
before 1	cut & remove 2	small slice scrap 3	hot glue 4
Most every antenna we did had this happen. This is what to do: Ref: 1 If you discover that the point between the “upper 1/2-wave element” and the “lower matching section” is starting to look like this image. Don’t begin to panic. No problems. Always solutions. Ref: 2 Just trim back a little of the insulation. About a 1/2 inch or whatever you decide works best. Ref: 3 Look through some of that scrap plastic that you removed and make yourself a little splint. Place a slice down one side. Now it can snap over the copper wire of the upper 1/2-wave element section. Ref: 4 Lock everything into place with a good coating of hot glue.			

SO-239 and Choke Modification:

450-Ohm Ladder Line is perfectly spaced for something like this.. This fun idea was a little time consuming to build. But it should make things a little more versatile. It's now ready for the hot glue treatment to cover over some of the exposed contact areas. There is still one open mounting hole on the SO-239 for the addition of a safety grounding system wire.





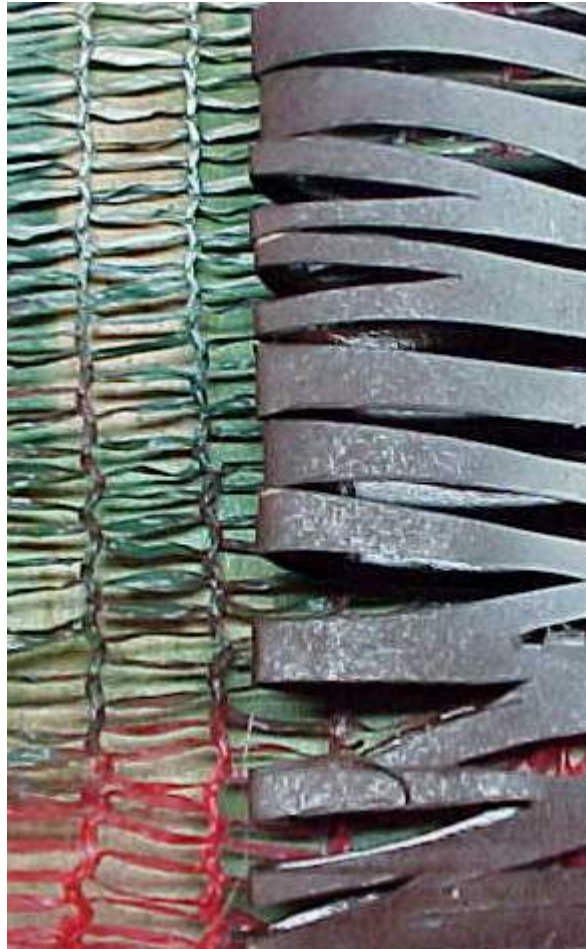
SWR results: 1.1:1 @ 146.000Mhz with this new configuration!
And this was without having to change any of the element lengths.
(..It's just one of the mysteries of antenna modeling..)

Stealth Mode Applications:(..And, GOOD LUCK finding this FOX..)

Black Ladder Line is just boring. Time for a facelift. **Kick it up a notch! Now you see it. But, not for long...BAM! What antenna? It's gone...** Anything will work. Just change the look. Punch holes or use your ideas. We used an old onion bag and a cardboard cutout for stenciling. **Use non-metallic based paints.**







You've got to have a little fun... No one said we had to be normal...Happy Fox Hunting!

Looking for a different project idea? This could be your lucky day!

Or Build a Phasing Coil!

(18 or 20 gauge solid copper wire works okay for this phasing coil...up to you.. Feed-point info for the coax hook-up stays the same..)

**2 Meter Collinear 1/2 Wave J-Pole
VHF-FM or SSB Antenna Design**
Made From
450-Ohm LADDER LINE

**Trying
this new
modification
idea can
almost double
your overall
antenna gain!**

**Feed Point
for the
50-Ohm Coax
now becomes
about
2 to 3 inches
from the
shorted end.**

Adjust for best
possible SWR
reading.

Upper
1/2 Wave
Element
Section

**39 and 1/2
inches**

Any length

(-)

Additional safety (-) ground.
Any length is okay..
Will work with or without..

Folded
1/2 Wave
Matching
Section

**18 and 1/2
inches**

Lower
1/2 Wave
Element
Section

**39 and 1/2
inches**

This is a dual purpose design!
Try a different mode!

Can be mounted vertically
for 2-meter FM operating.

But, also even horizontally
for 2-meter single sideband.

FM

SSB

